

Customizable Dashboards — Diagram Builder, Widget Builder and Customer-Defined Formulas

Field	Value
Product line	SandGen (Sand Generation), on a Sand Platform capability
Owner	Mohamed Moanes (Functional Lead / acting Product Manager)
Status	Draft
Date	2026-08-10
Skill used	sandgen-prd, with sandplatform-prd consulted for placement
Candidate plane (Sand fit)	plane:platform primary, with an explicit meta-plane seam to plane:agentsys for the chat assistant, surfaced through app:sandgen. Must be confirmed with architecture
Sand-fit depth	Full. Decided by the PRD Architect — see Confidence & Flags
Source brief	SB-20260809-customizable-dashboards-v2 (contract v2, Status: Ready for PRD, Content mode: PM-provided, PM confirmation 2026-08-09)
Declared scope (from brief)	Platform, applied to Sand Generation first. This PRD agrees on Platform for the composition and measure work and rejects the word "first" — part of this is permanently app-resident. See Sand Fit
Also carries	The dashboard composition material removed from the Admin Module PRD on 2026-08-09. That PRD, now at v4, no longer reproduces it — Requirements 22 to 28 are authored here and nowhere else
Document control	v2, 2026-08-10. Same conclusions as v1; reissued as two files on the output contract set 2026-08-10, drafted from brief v2, and re-grounded against the knowledge base. Nothing was deleted — every requirement, check, non-goal, risk, dependency and open question is in this document or the Annex. Revision record in the Annex
Companion	PRD-customizable-dashboards-v2-ANNEX — field dictionary, screen states, failure states, architecture reasoning, check methods, source record
Readers	Business / leadership · Product Owner · UI/UX Design · Engineering
Approver	[NEEDS INPUT: who signs this off – Functional Lead, CTO, or both]

Who should read what

You do not need all of it. Read your row.

If you are	Read	Skip
Business or leadership	Executive Summary, The Driver, Goals, Non-Goals, Pricing, Rollout	Everything technical
Product Owner	All of it	The Annex
UI/UX Design	Users & Scenarios, Design Deliverables, plus the field dictionary, screen states and failure states in the Annex	The architecture sections
Engineering	Requirements, Sand Fit, Data & Integration, Acceptance, plus the Annex	Pricing

Contents

Who should read what.....	2
Executive Summary.....	2
Assumptions	2
1. Problem & Motivation	2
2. The Driver	2
3. Goals & Success Metrics.....	2
4. Non-Goals / Out of Scope.....	2
5. Users & Scenarios.....	2
Which role model this uses.....	2
Who does what.....	2
Who may do what — this is the only copy	2
What people are trying to do	2
The main paths through it	2
When things go wrong.....	2
6. Requirements.....	2
6.1 Functional Requirements.....	2
6.2 Quality Requirements.....	2
6.3 Detailed Field Specifications	2
6.4 Design Deliverables.....	2
7. Sand Fit — Placement & Mechanism Reuse	2
Open placement questions for architecture.....	2
8. Multi-Tenancy, Apps & Composition	2
9. Data & Integration	2
10. Acceptance Criteria & Verification	2
11. Pricing / Packaging Impact	2
12. Risks, Open Questions & Dependencies	2
The one question filed as blocking	2
The risks worth acting on this week	2
What this waits on	2
Destructive-operation note	2
13. Rollout & Phasing.....	2
14. Decomposition Hint	2
15. Confidence & Flags	2
Open Questions	2

Executive Summary

What is being built. Three ways for a customer to build its own views in Sand Generation instead of only using the ones Sand ships, and one place to arrange them. A **Diagram Builder** for charts of the customer's choosing. A **Widget Builder** for panels showing the figures it cares about. A **Formula Builder** so a customer can define its own calculations — with a chat assistant proposing the formula and the person confirming it before it can be used. All three feed a dashboard the person arranges themselves, and any of them can be turned into a report.

Whose problem this is. The operators of generation assets — Asset Managers, O&M Managers and Engineers. Default dashboards with KPIs, widgets and diagrams have shipped and they work. What a customer cannot do is decide **which** chart, **which** figure or **which** calculation. When the view it needs is not one Sand ships, it exports the readings and works it out elsewhere, or asks a Sand person.

The driver, in one line. A **Pilot-stage** account is waiting on customizable dashboards before it moves to Full Onboarding, and Sand's own **Plus** package already sells *custom KPIs, charts and reports* that Sand cannot deliver today.

One thing the driver does not cover. Nothing anyone has said shows any account asked for a **chat assistant**. The outcome the Product Manager recorded — "*see any formula he wants*" — is met by a formula editor. The assistant is a separate, un-driven choice, kept separable here so leadership decides it on purpose.

What is explicitly not in scope. The Admin Module, which is its own PRD and which this work waits on. Customer-created roles. A Report Creator. An assistant anywhere else in Sand. Third-party or marketplace exposure. Sharing outside the customer's own account. Mobile. And — stated as a decision, not an omission — Sand does not ship a library of pre-written formulas in this release.

How success will be proven. Today the number of views customers build for themselves is zero, by definition. Success is an Asset Manager who wants a chart, a figure or a calculation Sand does not provide creating it in one sitting without asking Sand — proven by thirty-one checks, twenty-nine automatic and two needing a person to look at a screen.

The dates that matter. Sand's record of what has been built, last updated 2026-08-09, lists all three builders as **in progress now**. Their roadmap slot was planned for **Q1-2026, a quarter that has passed**; the current quarter is Q3-2026. The Admin Module this release waits on is expected **this month**. No dates are committed here, per the Product Manager's standing direction.

The decision being asked. Four things.

1. **Approve the scope and the four-stage shape.**
2. **Decide whether the chat assistant is wanted at all, or whether a formula editor is enough.** A leadership call. Stage 3b cannot be estimated, designed, procured or governed until it is answered.
3. **Settle with architecture whether a customer formula is a measure,** which blocks most of the formula work.
4. **Confirm which package this sells in.** Unlike the Admin Module, this capability sits on the far side of Sand's pre-defined-versus-custom line, and the tier it lands in has two names in Sand's own material.

Assumptions

These were inferred or carried across rather than confirmed here.

- **The Pilot-stage account and its ask are real and current.** Per Idea Brief SB-20260809-customizable-dashboards-v2, quoting the Product Manager on 2026-08-09. The account is identified by stage only. Sand's pipeline record notes its four-account table is incomplete, so this account may not be one of the four listed.
- **The builders described are the ones being built now.** Sand's record of shipped state lists the diagram, widget and formula builders as in progress. This document specifies what "finished" means for work already under way, which is why the acceptance ladder matters more than usual.
- **The Diagram Builder and Widget Builder screens exist in the running product at version 0.2.5.** Per the brief. Every field and control in the Annex's dictionary was read from shared screens by the upstream Brainstormer Project, not stated by the Product Manager, and is labelled accordingly.
- **Reporting can consume a saved view.** The Product Manager's own words say a customer can "*even create report with it*", and reporting shipped. Nobody has stated what connecting a saved view to a report involves.
- **The Admin Module's four roles are the operative role model** for who may build and who may only view.
- **The calculation engine that shipped is the right engine for customer formulas.** Whether they actually resolve through it is an open architecture question, not a settled assumption.

1. Problem & Motivation

Sand Generation ships default dashboards and they work. Across the different scopes, with their KPIs, widgets and diagrams, they shipped in August 2026 (09_shipped-state.md, last updated 2026-08-09). The platform underneath already pulls device data in and calculates KPIs automatically — roadmap slot P0.1, shipped.

The gap is not that nothing exists. The gap is choice. A customer cannot decide which chart it sees, which figure sits on its panel, or which calculation produces that figure. It gets Sand's selection.

That costs Sand in three ways.

It blocks a live account. Per the Idea Brief, a **Pilot-stage** account is waiting on both the Admin Module and customizable dashboards, and moves to Full Onboarding once they are finished. One account, identified by the stage that qualifies the evidence.

Sand sells this already. 01_company-and-product-overview.md lists SandGen **Plus** as including **custom KPIs, charts and reports**. Sand ships pre-defined ones. That is a capability named in a tier table that the product does not deliver — the same shape as the alerting exposure Sand has already recorded, where Basic sells *"alerts & ticketing"* against software that is built but not released.

Nobody has written down what finished means. All three builders are being built right now, against no specification. That is the most immediate cost, and it is why this document exists in the middle of a build rather than before one.

What a customer does instead, today, is not confirmed. The brief records the likely answer — accept the default, export the readings and work it out elsewhere, or ask a Sand person — and marks it [PROJECT-PROPOSED — not from PM] rather than observed. One conversation with an Asset Manager would settle it, and it is the cheapest open question here to close.

Two research findings shape how this is specified rather than whether it is built. A view offering a stretch of time but no choice of how values within it are combined can be silently misleading; and the performance ratio, a widely used industry figure, has several accepted definitions, so two engineers can produce different values for the same plant. Both are dealt with in Risks and answered by Requirements 2 and 16. **Neither is a reason to build, and neither is claimed as one.**

In plain words: the dashboards Sand ships are fine, but they are Sand's choices. A customer who wants a different chart, a different number or a different sum has no way to make one, and one customer is waiting.

2. The Driver

- ✓ **External** — a **Pilot-stage** account, quoted by the Product Manager on 2026-08-09, is blocked on this work. In the Product Manager's own words, with the account name removed at source:

"Our Client [account name removed] is currently waiting on both Admin Panel and Customizable dashboards to be finished so that each user can create his own dashboard and the customizable dashboard as a release is blocked by the admin module to be finished"

The same evidence carries the Admin Module PRD's external driver. One account is waiting on both, and the Product Manager confirmed on 2026-08-09 that one reason to build covers all three pieces here.

- ✓ **Structural** — **claimed here, where the brief declined it.** 01_company-and-product-overview.md lists SandGen **Plus** as including **custom KPIs, charts and reports**; Sand delivers pre-defined ones. A capability declared in a tier table and not delivered is a structural gap. **The brief tested only the entry package and correctly found no gap there; it did not test Plus.** One honest limit: Basic is what every SandGen customer holds, while nobody has confirmed a customer is on Plus today, so the commercial exposure is smaller than the alerting one. The gap in the declaration is real either way.
- ✓ **Generative** — **claimed narrowly, and it does not change the priority.** Once a customer can compose a view out of parts it chose, a class becomes expressible that cannot be expressed today: **customer-authored artefacts built from the same composition model.** The Report Creator is the named future member; customer-defined formulas are a second — a named, reusable, customer-owned computation other artefacts reference. The narrowness matters: **Sand's canvas and element composition substrate already exists**, so what is unlocked is *customer authoring over an existing substrate*, not new platform machinery. Claiming more would be over-claiming.

The classification is this document's, not inherited. The brief claimed External and explicitly declined Structural. This PRD upholds External, adds Structural on evidence the brief did not test, and claims the narrow Generative class the Admin Module PRD says leaves with this work. Three things the re-test checked, because in auto-draft mode nobody challenges this out loud: the account is named by a qualifying **stage** rather than a name; the two research citations are **not** the driver and

appear nowhere in it; and the brief's content mode is PM-provided, so an External claim is permissible — a Project-predicted brief claiming one would have been a contract violation.

The one place the driver does not reach. The account is waiting on *customizable dashboards*. **Nothing in the evidence shows anyone asked for a chat assistant.** The Product Manager's own description of the outcome — "*so the user can see any formula he wants according to his own preference*" — is satisfied by a formula editor. So the driver covers **charts, panels and customer-defined formulas**. It does **not** cover the assistant that writes them. That is not a reason to stop; it is why the assistant is a separately gated stage and the blocking question.

In plain words: one customer is stuck waiting, Sand's own price list already promises custom charts and KPIs, and building this once is what makes a report builder possible later. Nobody has asked for the chatbot part.

3. Goals & Success Metrics

This is a platform capability with an application surface, so the metrics mix one app-level adoption measure with the platform measures — reuse, and capability enabled.

Metric	Baseline → Target
Views a customer built for itself	0 today — measured, and zero by definition until this releases → [PROPOSED — not measured] at least one self-built view per person at the customer workshop. The Admin Module PRD proposed the same target before the work moved; nobody has agreed it
Sand-built views requested by ticket	[NEEDS INPUT: current requests per month, if anyone counts them] → zero for customers on the customizable tier
A second artefact type reusing the composition model without a second implementation	0 of 1 → 1 of 1. The Report Creator is the named candidate. This is the generative driver's own metric and it moves here from the Admin Module PRD with the work
Customer formulas resolving through the shipped calculation engine rather than a second calculation path	[NEEDS INPUT: unmeasured — depends on the placement decision] → all of them
Views that state how their values were combined	0 today — the current screens offer a period and a resolution but no combination choice → 100% of saved views and panels
Assistant-proposed formulas a person changed before confirming	No baseline exists → [PROPOSED — not measured] recorded from first release, as the only evidence Sand will have about whether the assistant is trusted or rubber-stamped

The first row is the honest one and the one to hold the work to. **The last row is deliberately given no target.** No independent research exists on how often people accept an incorrect formula written for them by an assistant — the brief searched and found only supplier marketing — so a target would be invented. Recording the number from day one is what makes the eventual decision evidence-based instead of a preference.

4. Non-Goals / Out of Scope

Not doing	Reason	Where it belongs instead
The Admin Module	Separated by the Product Manager on 2026-08-09. This release depends on it and does not contain it	PRD-admin-module-multi-users-v4, in progress, expected August 2026
Customer-created roles, and editing the predefined four	The four are enough to express "may build" and "may only view". Deferring costs nothing architecturally because every requirement here is written against a privilege, not a role name	A later creator-tools wave, per the Admin Module PRD's own non-goal
A Report Creator	Named as a future piece in the Admin Module PRD and never briefed. It is the <i>named consumer</i> of the generative driver here, which is why building it now would pre-empt the reuse this document is trying to prove	Its own brief and PRD. This release's job is to make it cheap, not to build it

Not doing	Reason	Where it belongs instead
An assistant anywhere else in Sand	The assistant here does one job — proposing a formula. A general assistant is a different capability with different governance, and Sand's architecture records genai governance floors it would have to be designed against	A separate PRD, if a driver ever appears
A library of Sand-authored formulas shipped with the product	A decision, not an omission. The moment Sand ships a formula called <i>performance ratio</i> , Sand owns its definition against an industry that has several	A commercial and product decision, deferred until the naming question is settled
Third-party or marketplace exposure of customer-built views	Ruled out for this work by the Admin Module PRD. Sand's public interfaces are in progress and the marketplace slot has not started	The Sand Platform roadmap — the public interface and marketplace slots
Sharing a view outside the customer's own account	Cross-account visibility is not a capability Sand has, and a view builder is the worst place to introduce it	Nowhere yet. It would need its own PRD and its own isolation argument
Mobile surfaces	The Sand Generation mobile app is in design, not built, and competes for the single designer	The mobile roadmap slot, once design capacity exists
Any change to the default dashboards that shipped	This release only adds. Nothing already live is altered or removed	A permanent limit, not a delay — and Requirement 28 makes it testable
Automatic selection of which chart suits which measure	Speculative. Nobody asked for it, and it is the archetype of unrequested cleverness in a builder	Not scheduled

Two of the brief's exclusions are not inherited as written. *Not a Report Creator* and *not user-created roles* arrived with no reason and no destination; both now have both. And the brief's *not open to outsiders* is split into two rows above, because third-party exposure and cross-account sharing fail differently and one of them is a tenancy risk rather than a scoping choice.

In plain words: charts, panels, formulas and one dashboard to arrange them on. No admin module, no custom roles, no report builder yet, no assistant elsewhere, nothing shared outside the customer's own account, and nothing already working gets changed.

5. Users & Scenarios

Which role model this uses

This document builds the permission table on the Admin Module's four roles — **Owner, Admin, Engineer, Technician** — **not on the eight commercial role names in `01\_company-and-product-overview.md`**. The reason is specific: this release ships behind the Admin Module, so the four are what the software will actually enforce when it lands. The eight — General Manager, Asset Manager, Business Manager, Finance Manager, O&M Manager, Regional Manager, Associate, Technician — describe who buys and uses Sand, which is why the personas in the problem section are Asset Managers, O&M Managers and Engineers. Only Technician appears in both lists. **No mapping between them is proposed here.**

Because the Admin Module also promises customer-created roles later, **every requirement below is written against a privilege, never a role name**. "The Owner may share" breaks the first time a customer renames a role. "A person holding *share a view* may share" does not.

Who does what

Persona	Authority	What they do with this capability
Owner	Highest	Everything below, plus deciding what the client's default dashboard is
Admin	Day-to-day administrator	Builds charts, panels, formulas and dashboards; shares them; deletes shared ones

Persona	Authority	What they do with this capability
Engineer	Technical analyst; reaches the Analysis area	The primary builder. The Analysis area is where the Diagram Builder opens from. Builds and shares within their granted sites
Technician	Field maintenance, view only, assigned sites only	Opens dashboards shared with them. Does not build
The Sand person who builds views today	Not a role here	Stops, once a client is handed over

Who may do what — this is the only copy

Permissions attach to **actions**, not role names. The assignment below is [ARCHITECT-PROPOSED – confirm] except where the Admin Module PRD already fixes it.

Action	Owner	Admin	Engineer	Technician
Build and edit a chart	Yes	Yes	Yes	No
Build and edit a panel	Yes	Yes	Yes	No
Build and edit a dashboard, and arrange it	Yes	Yes	Yes	No
Define, edit and delete a formula	Yes	Yes	Yes	No
Use the chat assistant to draft a formula	Yes	Yes	Yes	No
Share a dashboard with another person in the same client	Yes	Yes	Yes	No
Open a dashboard shared with them	Yes	Yes	Yes	Yes
Set the client's default dashboard	Yes — only role	No	No	No
Set their own default dashboard	Yes	Yes	Yes	Yes
Delete a dashboard someone else built	Yes	Yes	No	No
Build a report from a saved view	Yes	Yes	Yes	[NEEDS INPUT]

Everything in this table is bounded by site access, which this capability does not grant. A person who cannot see a site cannot chart it, cannot put it on a panel, cannot reference it in a formula, and does not see it when a dashboard containing it is shared with them. Site access is the Admin Module's to grant; this capability only ever reads it.

What people are trying to do

- As an **Engineer**, chart a measured value Sand does not chart by default, over a period and resolution I choose, so I can answer a question without exporting readings into a spreadsheet.
- As an **Engineer**, say whether my chart shows a total, a typical value or a peak, so the number on it means what I think it means.
- As an **Asset Manager working as an Admin**, define a calculation of my own and put it on a panel, so my dashboard shows my organization's figures rather than Sand's.
- As an **Admin**, share a dashboard with my team and have each person see only what they are allowed to see, so I do not build one dashboard per person.
- As a **Technician**, have the dashboard shared with me look designed for me, so I am not looking at a page of gaps and placeholders.
- As **anyone building a formula from other formulas**, be told before I save when I have made a loop, rather than after.

The main paths through it

A chart. Open the Diagram Builder from the Analysis area → name it → pick a scope and a measured value → choose a period, a resolution and how values are combined → choose a display style and a color → watch the preview redraw → save as a draft or properly → place it on a dashboard.

A panel. Same starting point → pick one of the four shapes → fill in a title, the measures and their units → choose the period and how values are combined → watch the preview → save → place it.

A formula. Open the Formula Builder → describe the calculation in ordinary words in the chat, or write it directly in the editor → the assistant proposes one and shows what it resolves to on real readings, for a scope and period the person picks → **the person confirms it** → it becomes available in the Widget Builder like any other measure.

A dashboard. Add sections; place charts and panels into them; move, copy, cut, paste and delete at any level, including pasting between two dashboards; share it; export it.

When things go wrong

Fifteen situations and the required behavior for each are in the Annex, authored there and nowhere else. They are requirements, not notes. The four that carry the most weight: a **shared dashboard** resolves what each recipient sees against their own access **at the moment they open it**, never at the moment it was shared; anything they may not see **degrades to a tidy placeholder** that does not break the layout; a **formula loop** is refused at save with the loop shown, never discovered at render time; and a **single-figure panel with no combination rule cannot be saved at all**, because a lone number with no statement of what it is a number of is the specific failure the research describes.

[Design-relevant] As authority drops, controls are **hidden, not greyed out** — and the layout must not leave a hole where a hidden control would have been. A Technician's dashboard should look designed for a Technician, not like an Engineer's with pieces missing.

6. Requirements

6.1 Functional Requirements

The Source column says where each requirement comes from. *PM* traces to the Product Manager's own words in the Idea Brief. *Admin PRD* is carried from the dashboard material removed from the Admin Module PRD on 2026-08-09, which is Product-Manager-reviewed. *Screens* is the upstream Project's reading of the running product, [PROJECT-PROPOSED — not from PM]. *New* is this PRD Architect's, and every one of those is [ARCHITECT-PROPOSED — confirm]. The reasoning behind each is in the Annex.

The Diagram Builder — charts

#	Requirement	Priority	Source	Proved by
Requirement 1	A person with the build privilege can create, preview and save a chart carrying a name, an optional description, a scope, one or more measured values, a period, a resolution, a display style and a color. The preview redraws against real readings as choices change	MUST / P0	PM	Check 1
Requirement 2	Every chart and every panel states how its values are combined — total, average, minimum or maximum — as an explicit choice, offered only where the measure's own type permits it. A view cannot be saved without one	MUST / P0	New	Checks 2, 3
Requirement 3	A saved view stores a default period and resolution. Whoever opens it can change the period without altering the saved view; saving the change is a deliberate second action	MUST / P0	New — resolves a contradiction	Check 4
Requirement 4	Display styles cover line, bar, area and points . Periods cover Day, Week, Month and Year; resolutions run from one minute to one day	SHOULD / P1	PM, plus a gap	Check 5
Requirement 5	A single control applies display style and color across all series in a chart at once, without removing per-series control	COULD / P2	Screens	Check 5
Requirement 6	A chart can be saved as a draft — visible only to its author and not placeable on a dashboard — or published	SHOULD / P1	Screens	Check 6

The Widget Builder — panels

#	Requirement	Priority	Source	Proved by
Requirement 7	A person with the build privilege can create, preview and save a panel in one of four shapes — a group of figures , several series together , a gauge , or a single figure — each carrying a title, its measures, their units and a tooltip	MUST / P0	PM	Check 7
Requirement 8	Every panel shape carries a scope and a period , not only one of them	MUST / P0	New — closes a gap	Check 8
Requirement 9	A single-figure panel cannot be saved without a combination rule and a stated period, and displays both alongside the number	MUST / P0	New	Check 3
Requirement 10	A gauge carries a target as well as a minimum and a maximum, and shows direction of travel against that target	SHOULD / P1	New	Check 9
Requirement 11	The panel builder has one name in the menu, on the page, on its save control and in every document. This PRD uses Widget Builder , the Product Manager's own word	MUST / P0	New	Check 10

Customer-defined formulas

#	Requirement	Priority	Source	Proved by
Requirement 12	A person with the define privilege can create a formula over measured values read from devices and over other formulas built from those readings , and use it in the Widget Builder like any other measure	MUST / P0	PM	Check 11
Requirement 13	A customer-defined formula resolves through the calculation engine that already ships , not through a second calculation path added for this feature	MUST / P0	New	Check 12
Requirement 14	A formula that would depend on itself — directly or through any chain — is refused at save , with the loop shown. A chain deeper than the permitted limit is refused at save with the limit stated	MUST / P0	New	Checks 13, 14
Requirement 15	Editing a formula names every saved chart, panel and dashboard that depends on it, and how many, before the change is saved. Deleting one is refused while dependents exist; deleting anyway is a second deliberate action after which dependents degrade to a placeholder rather than an error	MUST / P0	New	Check 15
Requirement 16	A formula carries a name, a description, a unit and its author. Where two formulas in one client account share a name, neither is ever rendered without something that distinguishes them	MUST / P0	New	Check 16
Requirement 17	Formulas are scoped to one client account. No formula, and no part of one, is readable from another client account by any route including the assistant	MUST / P0	New	Check 17
Requirement 18	The formula editor works without the assistant. A person can write, validate and save a formula directly. The assistant is one way into the editor, never the only way	MUST / P0	New	Check 18

The chat assistant

#	Requirement	Priority	Source	Proved by
Requirement 19	The assistant proposes ; the person confirms . Nothing the assistant produces is usable in the Widget Builder, on a	MUST / P0	PM	Check 19

#	Requirement	Priority	Source	Proved by
	dashboard, or in a report until a person has confirmed it. An unconfirmed proposal cannot be referenced by anything			
Requirement 20	At the moment of confirmation the person sees, together: the formula itself; a plain-language reading of what it computes; its unit; and what it resolves to on real readings for a scope and period they choose	MUST / P0	New	Checks 20, 31
Requirement 21	What the assistant is given, and whether anything leaves Sand, is declared in the product and settled before build. Until architecture rules otherwise, the assistant is given only the names, units and types of measures the person can already see — never readings, and never anything from another client account	MUST / P0	New	Check 21

The dashboard the three feed

#	Requirement	Priority	Source	Proved by
Requirement 22	A dashboard holds sections ; sections hold charts and panels ; the person arranges them	MUST / P0	Admin PRD	Check 22
Requirement 23	Copy, cut, paste and delete at every level — a single panel, a chart, a whole section, or an entire dashboard — including pasting between two different dashboards	MUST / P0	Admin PRD	Check 23
Requirement 24	A dashboard can be shared inside the client account. What each recipient sees is resolved against their own site access at the moment they open it , never at the moment it was shared. Anything they may not see degrades to a tidy placeholder that does not break the layout	MUST / P0	Admin PRD	Checks 24, 25
Requirement 25	Where a person lacks a privilege the control is hidden, not greyed out , and the layout leaves no hole where it would have been	MUST / P0	Admin PRD	Checks 26, 30
Requirement 26	A dashboard can be exported	SHOULD / P1	Admin PRD	Check 27
Requirement 27	A saved chart or panel can be used to build a report	SHOULD / P1	PM	Check 28
Requirement 28	The default dashboards that already shipped are unchanged by this release. Customer-built views are added alongside them	MUST / P0	New	Check 29

One thing this release deliberately does not add. The Admin Module already specifies a personal *default dashboard* preference limited to dashboards a person can reach. This release gives that preference customer-built dashboards to point at. It is not restated as a requirement here — it belongs to the Admin Module PRD and is named in dependencies instead.

In plain words: let people build their own charts and panels; make them say whether a number is a total, an average or a peak; let them define their own sums; never let a sum quietly depend on itself; always show what a proposed sum actually works out to before anyone accepts it; let people arrange all of it on a page and share it so each person sees only their own sites — and change nothing that already works.

6.2 Quality Requirements

The full quality-bar table is in the Annex. Four bars are specified and testable: every build, share and formula action is enforced server-side on the route as well as hidden on screen, with the API rung as the minimum because everything sits behind sign-in; client separation is structural and inherited from the Admin Module rather than invented here; **a view never widens what a person can see, and where site access and view access disagree, site access wins**; and new chart types, panel shapes and measure types are added by registering rather than by editing core.

Nine bars were never specified by anyone, and nothing was invented to fill them: preview responsiveness, reading volume per view, history depth, formula chain depth, whether anything must be live, assistant availability, assistant data handling beyond Requirement 21, accessibility, and retention. Each is [NEEDS INPUT] in the Annex and each appears in Open

Questions. Two deserve naming here. **Reading volume decides a product question, not just a performance one** — at one-minute resolution over a year a single series is over half a million points, and nobody has said whether that is a case to support or to refuse. And **accessibility is less optional here than in most Sand features**, because color is a primary carrier of meaning in every chart in this document.

6.3 Detailed Field Specifications

The complete field dictionary is in the Annex — chart, panel, formula and dashboard, with the unit and required-versus-optional conventions that apply to all of them. It is authored there and nowhere else.

One provenance point belongs in the reading path. Those fields were read from shared product screens **by the upstream Brainstormer Project, not stated by the Product Manager**, and they describe software that exists at version 0.2.5. They are a well-informed description, not a specification anyone signed, and **every row needs confirming against the running product before build.**

6.4 Design Deliverables

[Design-relevant]

Ten screens, four of them new from nothing.

Screen	New or extended	Who reaches it
Diagram Builder	Extended — exists at version 0.2.5	Build privilege
Widget Builder	Extended — exists at version 0.2.5	Build privilege
Formula editor	New	Define privilege
Formula chat assistant	New — nothing exists	Define privilege
Formula confirmation view	New , and the most safety-critical screen in this document	Define privilege
Formula library, with dependents	New	Define privilege
Dashboard canvas — arrange, copy, cut, paste	New	Build privilege
Dashboard share dialogue	New	Share privilege
Dashboard viewer, for a recipient	New	Anyone a dashboard is shared with
Chart and panel picker, when placing onto a dashboard	New	Build privilege

The states each screen needs — empty, loading, no permission, error, partly loaded — are set out per screen family in the Annex. **Every one is [ARCHITECT-PROPOSED — confirm]** and **none was specified by anyone**; the brief names surfaces, not states, and a designer cannot start without them.

The four things that need the most care.

- 5. The hardest interaction is the clipboard.** Copy, cut, paste and delete at four levels, including between two dashboards, is named in the Admin Module PRD as the hardest interaction in that whole module. Getting paste-target behavior subtly wrong is expensive to unpick later, and moving documents has not made it easier.
- 6. The second hardest is the confirmation screen.** A person is being asked to accept a calculation they may not be able to read. What they see there decides whether this feature is safe, and it is currently undesigned.
- 7. A shared dashboard is a normal case, not an edge case.** Most people will open dashboards someone else built, seeing a subset. Design for the partial render first and the complete one second.
- 8. Hidden, not greyed out — and no hole where the control was.**

Left to design's judgement: the canvas grid and drag affordances, how the clipboard is surfaced, the chat layout, the formula library's density, and how a chart preview shows a gap in readings.

Must NOT be invented: the four panel shapes, the four role names, the required-versus-optional split in the field dictionary, the unit-display rule, the confirm-before-use ordering, and the rule that controls are hidden rather than greyed.

Open design decisions. These go to design, not to architecture.

9. Every state in the Annex's matrix — proposed, unconfirmed.
10. Accessibility and breakpoint requirements, and specifically whether color alone may carry meaning in a chart.
11. How a chart distinguishes two formulas that share a name. Requirement 16 requires it; how it looks is design's.
12. Whether the formula library is a screen, a panel inside the Widget Builder, or both.
13. How a gauge shows direction of travel against its target, if Requirement 10 survives review.

In plain words: ten screens. Four are new and one of those — the screen where someone accepts a calculation — decides whether the whole formula feature is safe.

7. Sand Fit — Placement & Mechanism Reuse

Sand-fit depth: Full. That is this PRD Architect's decision: this specifies real behavior a developer implements against platform-dev, it reads device data, it composes rendered views, and it adds a new kind of computed value to the platform. A commercial-only treatment would be wrong.

Grounded in 08_sand-fit-architecture-digest.md, **generated 2026-08-02** — a dated snapshot of the platform-dev repository, not a live connection. **Every claim here is a candidate that must be confirmed with architecture.** Nothing came from the brief; briefs carry no architecture by contract and this one honored that. The full reasoning, the package table, the mechanism mapping and the architecture-decision table are in the Annex.

The declared scope, tested. The brief declared Platform level with Sand Generation first, and quoted the Product Manager. **Read that quotation carefully: it is about the Admin Module.** The upstream Project carried it forward as this work's scope declaration. That is a reasonable inference, it may well be what the Product Manager meant, and it is recorded as an inference rather than as a statement about dashboards.

This document agrees with Platform, rejects the word "first", and splits the work three ways.

14. **The composition and authoring substrate is platform, and already is.** Sand's canvas and element chain — ViewDocument → ViewCanvas → ViewElement, driven by CanvasRegistry, ElementRegistry and NodeTypeRegistry — is the platform's sixth fundamental mechanism, schema-driven dynamic views. Customizable dashboards register into it. **This inverts the usual objection:** placing a capability at platform level with one consumer is normally premature abstraction, but here the platform machinery already exists and building a bespoke renderer inside Sand Generation would be the speculative choice.
15. **Measures and formulas are platform.** scoping is the base vocabulary package and the digest names Measure and Quantity among the concepts it owns. A customer-defined formula is a new kind of measure, and it belongs where measures live.
16. **Part of this is permanently app-resident, not "first".** The Sand Generation shell, the Analysis area the builders open from, and the **catalogue of measured values a Generator customer sees** are Generator domain vocabulary. Sand Distribution will need a different catalogue over the same machinery. "Later it will be applied to any application with the same structure" is right about the machinery and wrong about the vocabulary, and confusing the two is how a second implementation gets built by accident.

In plain words: the parts that draw charts and hold sums are shared across all Sand products and already exist. The list of things a solar customer can chart is specific to solar and always will be.

Plane. Tag ``plane:platform`` as primary, with an **explicit meta-plane seam to ``plane:agentsys``** for the chat assistant, surfaced through app:sandgen. A PRD should tag exactly one plane or say it is a seam between two; this is genuinely the second case, and **the plane boundary is the safety boundary.** The assistant may only ever *propose*. The platform validates and persists, and only on a person's confirmation. An assistant that could write a measure directly would collapse that seam, and Requirement 19 exists to keep it.

In plain words: the charting machinery belongs to Sand's shared platform; the chatbot belongs to a different layer that is only allowed to suggest, never to save.

Reuse before invent — and this changes the size of the work. Rendering, styling, the element registry, the measure pipeline and the reading-frame abstraction all exist, the calculation engine shipped, and a saved view is **configuration data rather than a schema**, so new view types are data and not migrations. The Annex maps each one. **What this document actually adds is five things:** authoring interfaces, persistence of view definitions as configuration, customer-defined measures, the dependency safety around them, and the assistant. That is materially smaller than the name suggests, and it is why the formula placement question matters more than anything about charts. **Everything else is registration, with two boundary exceptions named in the Annex** and answered by the first and third questions below.

Open placement questions for architecture

Name	Question	Blocks the build?
Whether a customer formula is a measure	Is it a registered measure type resolved by the existing pipeline, or a separate kind of artefact? If it is a measure, Requirement 13 is nearly free, Requirement 2's combination options come from the measure's own type, and chaining is what two-level resolution already models. If it is not, Sand acquires a second calculation path	Yes — most of the formula work
Where a saved view lives	Configuration on an existing record under the JSON parameterization mechanism, or new tables?	Yes — it decides whether there is a migration at all
Whether the canvas substrate supports cross-document paste	Requirement 23 needs an element to move between two dashboards. Does the substrate express that today?	Yes — for the clipboard
What the assistant may see, and where it runs	Which plane, which supplier behind which adapter, and does anything leave Sand? Governed by Sand's recorded genai governance floors	Blocks the assistant stage only
What `packages/analytics` is	It appears in the digest's verified file locations but in none of its four package layers, and it is the likeliest home for half of this work	No — but it is half the placement argument

8. Multi-Tenancy, Apps & Composition

A platform capability with an application-specific surface — the configuration that most often needs an architect, and here it does. The composition substrate and the measure vocabulary are composed by every Sand application; the Analysis area, the shell and the Generator measure catalogue are Sand Generation's.

Separation is structural, inherited rather than invented. This capability creates no new separation mechanism. It relies on the Admin Module's model: a client identifier on every top-level record, a per-client sign-in space, and site access that cascades down the tree. Saved views, formulas and dashboards are client-scoped records like any other.

Three failures, in descending order of damage and ascending order of likelihood.

- A formula or view leaking data across client accounts.** Catastrophic and unlikely — and it acquires a new route in this release that did not exist before: **the assistant**. An assistant that learns from, or can retrieve, one client's formulas while helping another is a cross-account leak wearing a helpful face. Requirements 17 and 21 exist for exactly this, and it is why the assistant's inputs are declared rather than assumed.
- A shared dashboard showing a recipient a site they were never granted.** The realistic one. Prevented by resolving access at the moment the dashboard is opened rather than when it was shared — Requirement 24 — which is why that requirement is worded around *when*, not *whether*.
- A formula referencing a site its viewer cannot see, and quietly including it in a number.** The subtle one, and the easiest to miss in testing, because nothing looks wrong: the dashboard renders and the figure appears, computed over more data than the viewer is entitled to. **A number is as disclosive as a chart.** Check 25 treats it that way.

Composability. Because the capability registers into the platform's existing element and measure machinery rather than sitting beside it, a later artefact type — the Report Creator is the named candidate — consumes the same composition model without a second implementation. That is the generative driver's claim, and the third success metric is how it gets checked.

Cross-app visibility: none. Sand Distribution has not started. When it does, it composes the same substrate with its own measure catalogue.

In plain words: each customer's charts, sums and dashboards are walled off by the same structure that walls off their sites. The new risk this release adds is the chatbot, and it is the one to watch.

9. Data & Integration

The routing below comes from the Product Manager's **own words**, quoted verbatim in the Idea Brief. **The four sentences are reproduced in full in the Annex** because they are the only record of the original phrasing; the left column below quotes them verbatim, fragment by fragment, and every routing decision was matched against the quotation rather than a paraphrase of it.

What the Product Manager said	Where it routes
"readings from devices for the selected scope and selected period"	dataflow — DataAcquisitionService and the ReadingFrame family, over the time-series store. A scope, a window and a resolution is what a reading frame already is, so the Diagram Builder is an interface onto an abstraction that exists
"for the selected scope"	The polymorphic Scope hierarchy, bounded by hierarchical scope authorization. The builder never widens access
"plot it later on different time periods"	Not a storage concern — a view-time parameter. This is the sentence Requirement 3 resolves against the screens
"even create report with it"	The Reporting capability, shipped under roadmap slot G1.3. A saved view becomes a report input. What that connection involves is undocumented
"as Lines, or dots or area ... or bars" · "Gauge or group of parameters or single parameter or multi series"	Schema-driven dynamic views, the sixth mechanism; subject-display read-side elements for the figure shapes; element variants; the styling surface for display style and color
"the user custom dashboard/s" · people building their own views	The canvas and element composition model. The mapping table's own instruction applies — check for an existing registered type before calling anything new, which is what the reuse check did
"the formula ... from parameters readings from devices and other created formula "	The measure pipeline and the measure-typing-and-aggregation model. That clause is the whole architectural question: it means formulas stack, which is what two-level resolution exists to express
"An AI Agent will be available for the user to chat with"	Sand's recorded genai assistant architecture and genai governance floors, in plane:agentsys, with the model supplier behind an adapter
"after the user confirmation"	Not a data concern at all, and the most important five words in the quotation. It is the seam between the two planes, and Requirements 19 and 20 are built on it

In plain words for the table: device readings come from the part of the platform that already collects them, charts are drawn by the part that already draws them, sums run through the engine that already calculates KPIs, and the chatbot sits outside all of it and is only allowed to suggest.

Two things the Product Manager's words do not settle, and nobody else has either. Where the readings come from and how far back they go; and how many sites or devices one view may cover. Both are [NEEDS INPUT] and both are in Open Questions. Neither blocks the build, but the second decides whether a one-minute-resolution annual chart is a supported case or a refused one — a product decision hiding inside a performance question.

External systems. Only one is new: whatever supplies the assistant. It is reached through an adapter, and **Sand has no record of ever having chosen a supplier.** That is a procurement and governance decision, not an engineering one, and it belongs in the same conversation as whether the assistant is wanted at all.

Interface surface. New authoring routes behind the backend-for-frontend, which is why every check here is API-rung minimum. No third-party-facing interface is proposed.

Migrations — smaller than it looks. If saved views are configuration under the JSON parameterization mechanism, most of this release adds no tables at all. Formulas and dashboards likely need records; those are additive. **One operation is not purely additive:** editing or deleting a formula changes or breaks saved artefacts that depend on it — see the destructive-operation note.

10. Acceptance Criteria & Verification

Every check below is `[ARCHITECT-PROPOSED — confirm]`. Briefs carry no verification rungs by contract and no interview was run, so the whole ladder is this PRD Architect's proposal. Each states what it proves and the lowest sufficient rung.

Everything sits behind the proxy and the identity system, so the API rung is the minimum — a unit test proves nothing about a dashboard. **The method for each check is in the Annex.**

#	What it proves	Verifies	Verified at
Check 1	A chart saved with a scope, measures, period, resolution, style and color renders the same values on reload	Requirement 1	API
Check 2	A chart or panel cannot be saved without a combination rule, and the rules offered match what the measure's type permits	Requirement 2	API
Check 3	The same readings over the same window return different, correct values for total, average, minimum and maximum	Requirements 2, 9	CLI
Check 4	Opening a saved view with a different period returns data for the new period and leaves the saved default unchanged	Requirement 3	API
Check 5	Line, bar, area and points all render; the apply-to-all control changes every series	Requirements 4, 5	Playwright
Check 6	A draft chart is invisible to others and cannot be placed on a dashboard until published	Requirement 6	API
Check 7	Each of the four panel shapes saves and renders with its title, measures and units	Requirement 7	API
Check 8	Every panel shape requires a scope and a period	Requirement 8	API
Check 9	A gauge renders its target alongside its minimum and maximum	Requirement 10	Playwright
Check 10	The panel builder presents one name in the menu, on the page and on its save control	Requirement 11	Playwright
Check 11	A formula over measured values, and a formula over another formula, both resolve to correct values	Requirement 12	CLI
Check 12	A customer formula resolves through the shipped calculation engine, not a parallel path	Requirement 13	CLI
Check 13	A formula that would depend on itself, directly or through a chain, is refused at save and nothing is persisted	Requirement 14	API
Check 14	A chain deeper than the limit is refused at save with the limit stated	Requirement 14	API
Check 15	Editing a formula names its dependents before saving; deleting one is refused while dependents exist; forcing the delete degrades dependents to a placeholder	Requirement 15	API + Playwright
Check 16	Two formulas sharing a name are always rendered distinguishably — on a chart, a panel and a dashboard	Requirement 16	Playwright
Check 17	No formula is readable from another client account by any route, including the assistant	Requirement 17	API
Check 18	A formula can be written, validated and saved with the assistant unavailable	Requirement 18	API
Check 19	An unconfirmed proposal cannot be referenced by a panel, a dashboard or a report	Requirement 19	API
Check 20	The confirmation view shows the formula, its plain-language reading, its unit and its value resolved on real readings — and the confirm control is unavailable until the resolved value is present	Requirement 20	Playwright
Check 21	The assistant receives only what Requirement 21 permits, and any transfer outside Sand is what the product declares	Requirement 21	API
Check 22	A dashboard holds sections; sections hold charts and panels; arrangement survives reload	Requirement 22	API
Check 23	Copy, cut, paste and delete work at panel, chart, section and dashboard level, including pasting into a second dashboard	Requirement 23	Playwright

#	What it proves	Verifies	Verified at
Check 24	A recipient of a shared dashboard sees exactly what their own site access permits, evaluated at open time	Requirement 24	API
Check 25	A recipient cannot obtain data outside their sites through a shared view or through a formula on it	Requirements 24, 12	API
Check 26	Where a privilege is absent the control is absent from the response, not merely hidden by the screen	Requirement 25	API
Check 27	A dashboard exports, and the export contains only what the exporter may see	Requirement 26	API
Check 28	A report can be built from a saved chart and from a saved panel	Requirement 27	API
Check 29	The default dashboards that shipped render unchanged after this release	Requirement 28	Playwright
Check 30	A restricted person's dashboard reads as designed for them — no holes, no placeholder-only pages that should have said something	Requirements 25, 24	Operator-confirm
Check 31	A person who cannot read a formula can still tell, from the confirmation view alone, what it will do	Requirement 20	Operator-confirm

Traceability holds both ways. Every MUST and P0 requirement has at least one check, and every check names the requirement it verifies. The SHOULD and COULD requirements that could have gone unverified — the display styles, the apply-to-all control, the draft state, the gauge target, the export and the report path — have Checks 5, 6, 9, 27 and 28. **No requirement is unverified.**

Both operator-confirm checks gate close, not dispatch. Check 30 needs Checks 24 and 26 to pass first; Check 31 is the check that decides whether the assistant is safe to ship. Neither substitutes for a lower rung.

Three checks cannot be executed yet, and that is a real gap. Check 12 waits on the formula placement question. Check 14 waits on a depth limit nobody has set: [NEEDS INPUT: the limit]. Check 21 waits on the assistant's data-handling answer.

In plain words: twenty-nine of the thirty-one run automatically. Two need a person to look — whether a restricted dashboard reads as designed, and whether someone who cannot read a formula can still tell what it does — and both happen last.

11. Pricing / Packaging Impact

Tier: SandGen Plus — and unlike the Admin Module, this one genuinely sits on the far side of Sand's pre-defined-versus-custom line. Per 01_company-and-product-overview.md, Basic includes *pre-defined KPIs, charts and reports* and Plus includes *custom KPIs, charts and reports*. All three pieces here cross that line: the Diagram Builder makes custom charts, the Widget Builder makes custom KPI panels, and the Formula Builder makes custom KPIs outright. 01 also places *Web + Mobile* in Plus; this release is Web only, which is a partial delivery of the tier rather than a conflict.

Pricing rows used. 05_pricing-packaging-competitive.md's per-MWp SandGen table, quarterly and annual, across all four zones. Quarterly, **Basic** runs \$101 to \$187 per MWp and **Advanced** runs \$116 to \$216, depending on zone and installed scale. Annually, **Basic** runs \$350 to \$650 and **Advanced** \$403 to \$750. The gap — roughly **\$15 to \$29 per MWp per quarter**, or **\$53 to \$100 a year** — is what Sand charges for custom reports and billing today, and this capability is a large part of what that gap is meant to buy.

No bracket shift. Nothing here changes installed capacity or zone, so no customer crosses the sub-1 MWp, 1-to-100 MWp or 100-MWp-plus boundaries, and the hybrid and micro-grid ten-percent add-on is unaffected.

The tier-name ambiguity, and why it matters more here than in any Sand PRD so far. 01 packages SandGen as Basic / Plus / Gen Add-Ons; 05 prices it as Basic / **Advanced**, glossed as "*Custom Reports & Billing*". **Nothing in Sand's material states that Plus and Advanced are the same tier.** The Admin Module PRD could set this aside because it relied only on Basic, which appears in both. **This document cannot** — it relies entirely on the tier with two names and one price row, whose gloss mentions custom *reports* and *billing* but not custom *charts* or *KPIs*. So the priced tier may not be the packaged tier, and what this document specifies may not be inside either as written. **Not resolved here.**

Three things Commercial should know.

20. **Plus already sells this.** That is the structural driver restated as a commercial fact.
21. **The waiting account's tier is unknown.** Whether it is on Basic or Advanced decides whether shipping this closes a promise or gives away a tier. [NEEDS INPUT].
22. **The chat assistant has no tier at all.** No Sand pricing or packaging material mentions an assistant of any kind. If it ships, it is an unpriced inclusion in Plus, a Gen Add-On, or something new — and that is a decision, not a detail.

12. Risks, Open Questions & Dependencies

The one question filed as blocking

Whether the chat assistant is wanted at all, or whether a formula editor is enough.

It is filed as *the* blocking question rather than an architecture item deliberately. Everything else can proceed while it is open: Stages 1 and 2 do not touch it, and Requirement 18 makes Stage 3a shippable without it. But **Stage 3b cannot be estimated, designed, procured or governed until it is answered**, and it is the only question here that no engineer and no architect can answer. It belongs to leadership, and the evidence for it is thin on purpose: the account is waiting on customizable dashboards, not on a chatbot.

Two architecture items also block the build — whether a customer formula is a measure, and where a saved view lives — and they are carried in the must-confirm-with-architecture list rather than filed as questions, because they go to a different person and block a different thing.

The risks worth acting on this week

The full risk table is in the Annex. These are the ones that change what someone does now.

Risk	What to do
The work is already being built, against no specification. Sand's record lists all three builders as in progress	Check what is being built against the acceptance ladder — especially Requirements 2, 8, 14 and 19, none of which appear to be in the screens as they stand
A silently wrong number. A view offering a period and a resolution but no combination rule reads as a total, a typical value or a peak depending on a choice nobody made	Requirements 2 and 9 make the choice explicit and required; Check 3 proves the arithmetic. The highest-value change in this document
Two figures with one name. A customer names a formula <i>performance ratio</i> ; the industry has several accepted definitions; two dashboards disagree and nothing says why	Requirement 16 requires them always to be distinguishable and Requirement 16's mandatory description gives a reader something to distinguish them by. Who owns the name is a commercial question and stays open
The assistant is built because it was described, not because it was chosen	It is a separately gated stage, Requirement 18 keeps the editor independent of it, and the decision is filed as the blocking question
The assistant becomes a cross-account route	Requirements 17 and 21, and Check 17's second half, which asks the assistant in one client account about a formula that exists only in another
Nobody has chosen an assistant supplier, and no roadmap slot covers one	Treat supplier selection as a procurement and governance task with its own lead time, not an engineering choice made during build
One designer, and the mobile app is in design. Ten screens, several new from nothing	Decide explicitly which queue wins. The formula confirmation view is the one screen that cannot be deferred, because it is the safety of the whole third stage

What this waits on

Checked against 09_shipped-state.md (last updated 2026-08-09) first, then 06_roadmap-team-financials.md. **The full dependency table and the rejected alternatives are in the Annex.**

This is not new work. Roadmap slot G1.6 is **partial** — the defaults shipped, the customizable builders are in progress — and its planned quarter, **Q1-2026, has passed**. This document is a change of scope to work already under way.

Shipped and available: data acquisition with the calculation engine, the unified data model, the default dashboards, reporting, and the revamped user interface. Requirements 13, 27 and 28 all depend on shipped work. **In progress, and not to be treated as available:** the **Admin Module**, which the Product Manager states holds up this release and which every action

in the permission table depends on. **Genuinely unknown:** the developer toolkit, which sits with Sand's System Architect — who the Product Manager is separately waiting on for the design of the third piece.

Absent entirely: an assistant capability of any kind appears nowhere in the roadmap, the shipped record or the out-of-roadmap table. **It is the one genuinely new thing in this document** — though Sand's architecture does record genai decisions it must be designed against, which the brief could not have known.

Destructive-operation note

One operation is flagged. Editing or deleting a customer-defined formula changes or breaks every saved chart, panel and dashboard that depends on it — including artefacts belonging to other people in the same client account. It is not a data migration, but it is destructive to existing customer work, and it needs the three-part discipline: **an authorized scope** naming exactly which dependents are affected, **preserved prior state** so an edit can be undone, and **gates before and after** that count what changed. Requirement 15 is that discipline expressed as behavior and Check 15 proves it.

Everything else is additive. If saved views are configuration under the JSON parameterization mechanism, most of this involves no migration at all.

13. Rollout & Phasing

Additive first, and unusually literally so. The default dashboards that shipped stay exactly as they are (Requirement 28). Customer-built views are added alongside them. Nothing is removed, and the Sand-person-builds-it-for-you path keeps working throughout — retired only after at least one client has built and shared its own dashboards successfully, never before.

All phasing below is ``[ARCHITECT-PROPOSED — confirm]``. Rollout is never settled by a brief, and no interview was run.

Stage	What is in it	Can it ship alone?
1 — Charts that mean what they say	The Diagram Builder complete: build, preview, save, place on a dashboard. The combination rule made explicit and required. The saved-default period. Display styles including points. Draft and published states. <i>(Requirements 1 to 6, 22, 28)</i>	Yes — the named shippable intermediate state. It is also the smallest thing that satisfies the driver in part, and it carries the combination-rule fix, which is the highest-value item here
2 — Panels, and the dashboard that holds everything	The Widget Builder's four shapes with scope, period and combination rule on all of them; the gauge target; one name for the builder. Then the composition work: sections, the clipboard at four levels including cross-dashboard paste, sharing resolved at open time, graceful placeholders, hidden-not-greyed, export, and the report connection. <i>(Requirements 7 to 11, 23 to 27)</i>	Yes — and the end of Stage 2 is where the external driver is satisfied in full
3a — Formulas, editor first	Define a formula over measures and over other formulas; resolution through the shipped calculation engine; loop and depth refusal at save; dependency safety on edit and delete; naming and separation; the formula library. <i>(Requirements 12 to 18)</i>	Yes. This is the stage that delivers the Product Manager's stated customer outcome — <i>"see any formula he wants"</i> — with no assistant involved
3b — The chat assistant	Propose-and-confirm, the confirmation view, and the declared data handling. <i>(Requirements 19 to 21)</i>	Yes, and gated. It does not start until the blocking question is answered and a supplier and governance position exist. If the answer is "an editor is enough", this stage does not happen and nothing else in this document changes

Why the clipboard sits in Stage 2 rather than Stage 1, against the Product Manager's stated build order of diagram, then widget, then formula. That order is kept for the *builders*. The composition work — the clipboard, sharing, degradation — is the piece the Admin Module PRD named as the most complex part and planned to ship last and separately, precisely so it could not drag everything else with it. Putting it beside the Widget Builder keeps that protection while preserving the Product Manager's sequence.

Start narrow. Enable for the Pilot-stage account first, then widen. No feature-flag approach was specified; if one is wanted, that is an open item rather than an assumption made here.

No dates are committed, per the Product Manager's standing direction. The roadmap slot's planned quarter has passed and is stated as passed rather than restated as a target.

Sizing is not estimated. Two open things would change any estimate materially: whether a customer formula is a measure the platform already knows how to resolve, and whether the canvas substrate already supports moving an element between two dashboards.

14. Decomposition Hint

A starting point for Sand's planning workflow, not a breakdown.

- **Candidate epic title:** Customer-authored views for Sand Generation — diagram builder, widget builder, customer-defined formulas, and the dashboard that composes them.
- **Suggested priority: P0.** Justified against a named External driver — a Pilot-stage account blocked from Full Onboarding — and a Structural driver confirmed independently against 01_company-and-product-overview.md. **Stage 3b alone is P2**, because no driver reaches it.
- **Plane tag:** plane:platform with a meta-plane seam to plane:agentsys, surfaced through app:sandgen, per Sand Fit.
- **Skill hints, where the digest supports them:** skill:mech-schema-driven-views, skill:mech-registry-extension, skill:mech-json-parameterization, skill:pkg-dataflow, skill:pkg-scoping, skill:adr-035, skill:adr-026, skill:adr-029, skill:adr-024. **No skill hint is offered for the canvas-element-substrate or measure-typing decisions, because both are numbered ADR-031 and the collision is unresolved.**
- **Acceptance class:** accept:autonomous for Check 1 through Check 29; accept:operator-confirm for Check 30 and Check 31.
- **What the first task should be: not a builder screen.** Answer whether a customer formula is a measure the existing pipeline resolves, and get a decision on the assistant. The first makes Stage 3a cheap or expensive; the second decides whether Stage 3b exists. Both are answerable this week and both already block work that has started.

15. Confidence & Flags

The full record is in the Annex — what is solid versus assumed, the age of every source, the known ambiguities and how each was handled, the brief and Admin-Module provenance, every overrule, the upstream hypotheses relied on, the seam notes, the auto-draft disclosure and the marker record. Restated here is only what someone has to act on.

Must be confirmed with architecture, by name. Three block the build.

- **Whether a customer formula is a measure** the existing pipeline resolves — **blocks most of Stage 3a.**
- **Where a saved view lives** — configuration, or new tables — **blocks the schema decision.**
- **Whether the canvas substrate supports cross-document paste** — **blocks the clipboard in Stage 2.**
- **What the assistant may see, where it runs, and whether anything leaves Sand** — blocks Stage 3b only.
- **What `packages/analytics` is, and which layer it sits in** — half the placement argument rests on it.
- **Both files numbered ADR-031**, canvas element substrate and measure typing, are load-bearing here.
- Every package, mechanism and decision named in Sand Fit and Data & Integration is a candidate.

Must be confirmed with the Product Manager, leadership or Commercial.

- **Whether the chat assistant is wanted at all.** The blocking question.
- **The permitted depth of a formula chain**, without which Check 14 cannot run.
- **Which package this sells in**, given that the tier it belongs to has two names and one price row.
- **The name of the panel builder**, currently three.
- **Whether separating the documents also separates the releases**, which nobody has decided.
- **Who owns the name of a formula** when a customer's differs from Sand's.

Snapshot caveat. Architecture grounding comes from 08_sand-fit-architecture-digest.md, **generated 2026-08-02** — a dated snapshot of platform-dev, not a live connection, recording a live duplicate-ADR-031 numbering collision as a standing reminder that decision numbers drift. **Both halves of that collision are load-bearing here**, which is the first time that has happened, so every architecture-decision reference must be confirmed by filename against the live repository. The genai decisions that govern the assistant are named in the digest as **topics with no verified numbers**, so their numbers must be looked up fresh and never inferred from the digest's list order — the digest records a downstream PRD getting exactly that wrong on 2026-08-02. Commercial grounding comes from 01_company-and-product-overview.md and 05_pricing-packaging-competitive.md, both **September-2025** snapshots. 09_shipped-state.md is human-maintained, was **last**

updated 2026-08-09, and wins over the older roadmap wherever it speaks. **The live repository and the current price sheet are the final authority, not this document.**

Known source ambiguities touched, none resolved silently. Three: the two tier vocabularies, dealt with in Pricing rather than picked between; Sand's two live role lists, where the permission table uses the Admin Module's four and no mapping was invented; and the duplicated ADR-031, which is why decisions are cited by filename throughout. The Annex gives the handling for each.

Open Questions

Architecture

- Is a customer formula a registered measure the existing pipeline resolves, or a separate artefact? — **blocks most of Stage 3a**
- Does a saved view live as configuration, or in new tables? — **blocks the schema decision**
- Does the canvas element substrate support moving an element between two documents? — **blocks the clipboard**
- Where does the assistant run, which supplier sits behind the adapter, and does anything leave Sand?
- What is packages/analytics, and which package layer does it belong to?
- Which of the two files numbered ADR-031 governs which half of this work?
- Where do readings come from, how far back do they go, and how many sites may one view cover?
- Does anything on a dashboard have to be live rather than refreshed?
- What does building a report from a saved view actually involve?

Product and design

- Every state in the Annex's screen-state matrix — proposed, unconfirmed.
- Accessibility and breakpoints, and whether color alone may carry meaning in a chart.
- What is the panel builder called? Requirement 11 requires one name; which one is the PM's choice.
- Does a gauge need a target? Proposed from research, with no Sand evidence behind it.
- How does a chart distinguish two formulas that share a name?
- Is the formula library a screen, a panel inside the Widget Builder, or both?
- Is a feature-flag rollout wanted, or is per-client enabling enough?

A user or customer

- What does an Asset Manager do today when the view they want is not one Sand ships?
- Which of the four panel shapes do people actually reach for?
- **Three real examples of calculations someone would ask for.** — the highest-value gap in this document

Leadership and commercial

- **Is the chat assistant wanted, or is a formula editor enough? — blocking**
- Which package does this sell in, given that Plus and Advanced are not documented as one tier?
- Which tier is the waiting Pilot-stage account on?
- Does the chat assistant have a tier at all? No Sand pricing material mentions an assistant.
- Who owns the name of a formula when a customer's *performance ratio* differs from Sand's?
- Does separating this document from the Admin Module PRD also separate the releases?
- Should the generative reason claimed by the Admin Module PRD be formally restated, now this document claims it?
- Which queue wins for the single designer — these ten screens or the Sand Generation mobile app?

Data nobody has

- Nothing measured exists about how customers use the default dashboards or Sand's KPIs.
- The permitted depth of a formula chain.
- Preview responsiveness, view scale, history depth, retention and accessibility targets.
- How many self-built views a customer would need before this counts as adopted.